

OpenACC Installation Instructions

Note

GPU acceleration with OpenACC is not natively supported on Windows, macOS, or AMD GPUs. Full OpenACC support requires a Linux system with an NVIDIA GPU.

Step 1: Check for NVIDIA GPU

Open a terminal and run:

```
1 lspci | grep -i nvidia
```

If an NVIDIA GPU is available, it will appear in the output.

Step 2: Install NVIDIA Drivers

Check if NVIDIA drivers are installed.

```
1 nvidia-smi
```

If drivers are already installed, the above command will display the GPU details. Proceed to Step 3.

If it does not print any output, install NVIDIA drivers using the commands below.

```
1 sudo apt update
2 sudo ubuntu-drivers autoinstall
3 sudo reboot
```

Verify installation:

```
1 nvidia-smi
```

Step 3: Install NVIDIA HPC SDK

For Ubuntu:

```
1 curl https://developer.download.nvidia.com/hpc-sdk/ubuntu/DEB-GPG-KEY-NVIDIA-  
   HPC-SDK | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-hpcsdk-archive-  
   keyring.gpg  
  
3 echo 'deb [signed-by=/usr/share/keyrings/nvidia-hpcsdk-archive-keyring.gpg]  
   https://developer.download.nvidia.com/hpc-sdk/ubuntu/amd64 /' | sudo tee /etc/  
   apt/sources.list.d/nvhpc.list  
  
5 sudo apt-get update -y  
  
7 sudo apt-get install -y nvhpc-26-1
```

For other Linux distros:

```
1 wget https://developer.download.nvidia.com/hpc-sdk/26.1/  
   nvhpc_2026_261_Linux_x86_64_cuda_13.1.tar.gz  
  
3 tar xpf nvhpc_2026_261_Linux_x86_64_cuda_13.1.tar.gz  
  
5 nvhpc_2026_261_Linux_x86_64_cuda_13.1/install
```

Step 4: Configure Environment Variables

Add the compiler path to the environment:

```
1 echo 'export PATH=/opt/nvidia/hpc_sdk/Linux_x86_64/25.1/compilers/bin:$PATH' >>  
   ~/.bashrc  
2 source ~/.bashrc
```

Step 5: Verify Installation

Check compiler availability:

```
1 nvc --version      # C compiler (or) pgcc  
2 nvc++ --version   # C++ compiler (or) pgc++  
3 nvfortran --version # Fortran compiler (or) pgfortran
```

If successful, version information will be displayed.